Real-Time Code-Mixed Hindi-to-English Speech-to-Speech Translation with Causal Alignment and an 80 ms Front End

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Alignment audit: 42.6 to 14.8 ms MAE | Front-end: 1.84-3.01 ms per 80 ms frame

Abstract

Live Hindi-to-English speech translation must emit audio before the speaker finishes while protecting names, Indian monetary units, negation, and spoken repairs. This paper consolidates a candidate architecture and preliminary signal-processing results for the Sarvam.ai qualifying mini problem statement. The design uses a strictly causal 80 ms audio front end, forced-alignment boundary clamping, a bypass gate for clean or telephone-band audio, and a proposed dual-target data strategy. On one 25.47 s, 72-word code-mixed recording, a 30-boundary manual audit reports a reduction in alignment mean absolute error (MAE) from 42.6 ms to 14.8 ms (65.3%). Across five degraded versions of that clip, the reported front end takes 1.84-3.01 ms per 80 ms frame. Un-gated DeepFilterNet3 degrades SI-SDR in all five conditions, including -2.42 dB on clean telephone-band speech. These are preliminary, single-clip experiments: they do not establish end-to-end translation quality or generalization.

1 Introduction

The target setting is streaming speech-to-speech translation from naturally code-mixed Hindi to English. Incoming audio is mono 24 kHz PCM, delivered in 80 ms frames; translated English audio is emitted while the caller is still speaking. Since emitted audio cannot be retracted, a system must trade off low latency against the risk of committing to a value, name, verb, or negation before the Hindi utterance provides enough evidence. Financial phrases such as “one crore twenty lakh” and mid-utterance repairs such as “ten lakh ... no, wait ... eight lakh” are especially consequential. Simultaneous translation research similarly treats latency and translation quality as coupled objectives [1].

This submission combines a proposed data engine and repair-aware translation strategy with measured tests of audio alignment and denoising. We distinguish the implemented signal-processing evaluation from components that remain design proposals, and report timing as front-end timing rather than claiming a benchmark for a trained end-to-end speech-to-speech model.

2 System Design

2.1 Causal streaming contract

At 24 kHz, each 80 ms input frame contains 1,920 samples (3,840 bytes of signed 16-bit mono PCM). Processing is chunk-by-chunk with no future lookahead. The front-end design carries filter and overlap-add state across frame boundaries and includes stateful resampling, a causal Wiener filter, a causal DeepFilterNet3 path, Silero voice-activity detection (VAD), and a SmartCausalGate that can bypass aggressive filtering on clean wideband or telephone-band input. VAD uses a 240 ms causal hangover to avoid cutting short inter-phrase transitions.

Streaming front-end (strictly causal; no future lookahead)

Persistent state crosses frame boundaries; latency reported for this front-end only.

Figure 1. Conceptual front-end path. The reported 1.84-3.01 ms latency is for this front end, not a full translation-to-audio session.

2.2 Alignment and repair-aware targets

Hindi transcripts are normalized while retaining a verbatim form. MMS forced alignment (MMS_FA) supplies word boundaries; the proposed RMS-energy clamp adjusts boundaries near low-energy gaps to reduce leakage into hesitation silence. For training, the plan stores two English targets: a literal stream that preserves conversational repairs, and a fluent stream that resolves the final intent. A causal constraint requires an English word to start no earlier than its dependent Hindi word ends plus a safety margin. In a repaired amount, “eight lakh” must be aligned to the corrected “eight lakh” span, not to an earlier abandoned “ten lakh” span.

A proposed Prosody- and Entropy-Gated Speculative Hold (PEG-SH) would normally translate with a small lag, but hold a high-risk numeric or verb token for up to 400 ms when a post-entity pause or high decoder entropy suggests a likely repair. If the repair arrives after audio was committed, the literal strategy translates the repair explicitly rather than silently replacing the earlier value. PEG-SH is a proposal, not a measured result in the reported benchmark.

3 Dataset Engine (Proposed)

The proposed 200-hour Part A budget allocates 75 hours to real speech and 125 hours to targeted synthetic speech. The real portion is planned as 55 h IndicVoices, 10 h IndicVoices-R, and 10 h Kathbath; synthetic targets would use a single voice per unique script to avoid spending hours on duplicate multi-voice renderings. IndicVoices provides broad natural speech coverage [2]. This split is a recommendation, not evidence that the complete 200-hour corpus has already been assembled.

Each candidate record is intended to pass format and clipping checks, ASR round-trip filtering (target CER <= 5% and WER <= 10%), alignment confidence and timing checks, translation quality filtering, and deterministic checks for numbers, names, and negation. Proposed on-the-fly noise and codec augmentation does not add recorded duration to the 200-hour budget. The entity checker is useful as a guardrail, but string checks alone cannot reliably detect morphological negation (“unsigned”) or paraphrased refusals (“withhold payment”).

4 Experimental Setup

The reported test bed is a Windows 11 laptop with Intel Core i7-14650HX (16 cores/24 threads) and NVIDIA RTX 5060 Laptop GPU (8 GB VRAM). Six natural code-mixed clips were recorded; the featured negotiation clip is 25.47 s long and contains 72 words. The preprocessing pipeline (Whisper large-v3-turbo, normalization, MMS_FA with RMS clamping, dual translation, and entity checks) is reported to process this clip in 21.97 s with 1.80 GB peak VRAM. This is an offline preprocessing time, not an end-to-end streaming latency result.

Five degraded conditions were generated from the featured clip: babble plus fan/hum at nominal SNRs of 15, 10, and 5 dB; telephone-band audio limited to 300-3,400 Hz, downsampled to 8 kHz and G.711 mu-law quantized; and telephone-band audio with additional 10 dB noise. SI-SDR is computed against the clean 24 kHz reference after delay alignment. The accompanying RESULTS.md documents the conditions and reproduction commands; no independent rerun is claimed here.

5 Results

5.1 Word-boundary audit

A manual Audacity audit of 30 boundaries (15 starts and 15 ends) in the featured clip reports raw MMS_FA MAE of 42.6 ms, compared with 14.8 ms after 5 ms RMS-energy clamping. This is a 65.3% reduction, but the small, single-clip audit should be treated as preliminary rather than a population-level error estimate.

Condition	Input	Wiener	DFNet3	Smart gate	ms/frame
Babble + hum, 15 dB	+14.20	+14.84	+6.82	+13.51	2.24
Babble + hum, 10 dB	+9.21	+9.91	+5.12	+9.46	2.08
Heavy noise, 5 dB	+4.22	+4.88	+2.67	+4.57	1.84
Telephone band	+1.13	+1.13	-1.29	+1.07	2.21
Telephone + noise	+0.05	+0.57	-5.14	-0.05	3.01

Table 1. Delay-aligned SI-SDR (dB) for the degraded input and each processed output; final column is reported latency per 80 ms frame. Delta values are discussed in the text. Source: RESULTS.md.

5.2 Denoising outcomes and runtime

Classical Wiener filtering improves SI-SDR by 0.51-0.70 dB in the four noisy conditions and is neutral on the clean telephone-band condition. Un-gated DeepFilterNet3 lowers SI-SDR in all five conditions; the largest reported drop is 7.38 dB at 15 dB babble/hum, and the telephone-band drop is 2.42 dB. The Smart-gated path stays much closer to the input than the un-gated neural denoiser, but its SI-SDR is not uniformly better than the input or the Wiener baseline.

The Smart-gated front end is reported to process a frame in 1.84-3.01 ms, corresponding to a per-frame real-time factor of 0.023-0.038 (about 26-43 times faster than real time). These figures apply only to the audio front end. The report also describes Whisper failures on over-denoised word slices around “Sector 62”, “ten lakh”, and the corrected “eight lakh”; those are illustrative error cases, not a statistically powered ASR evaluation.

5.3 Entity and repair checks

The supplied report states that the deterministic entity checker passed the standard literal and fluent targets across six clips, including the name Rahul, Sector 62, the initial 1.20-crore price, and the final 8-lakh discount. In fluent mode, the abandoned 10-lakh amount is intended to be removed. The authors also report adversarial failures: morphological negation such as “unsigned”, lexical paraphrase such as “withhold payment”, and multi-clause repairs where both values appear without an explicit repair marker. Therefore, the reported 100% is a pass rate on the small standard test set, not a general entity-accuracy estimate.

5.4 Word-level failure cases

The supplied benchmark includes three targeted word-slice examples intended to show why waveform enhancement must be assessed with language-level checks. For the 18.50-22.00 s repair span (“nahin ruko ... aath lakh”), the reported segment SI-SDR is +0.49 dB for the degraded signal, -1.50 dB after un-gated neural denoising, and +0.48 dB after Smart-Gated processing. Whisper reportedly produces gibberish on the over-denoised segment, while the gated output preserves the 10-versus-8-lakh correction in the transcript. The close SI-SDR values of the degraded and gated waveforms are consistent with a conservative bypass, while the transcription contrast illustrates that a small waveform change can have a large lexical consequence.

For the 15.80-16.45 s “das lakh” segment, the report gives +1.38 dB on the degraded audio, +1.14 dB after neural denoising, and +1.34 dB after Smart-Gated processing. The denoised ASR output reportedly repeats “kar do”, whereas the gated output recognizes “to das lakh”. For the 6.50-7.20 s “baasath” segment, reported SI-SDR falls from +3.88 to +2.78 dB with neural denoising and is +3.65 dB with gating; the denoised ASR output reportedly loses the number 62. These are selected examples from one recording, not estimates of average word error rate.

5.5 Reading the benchmark correctly

The input and output SI-SDR values answer a signal-fidelity question relative to a paired clean reference. They do not, by themselves, measure speech intelligibility, transcription accuracy, translation adequacy, or listener preference. A method can preserve global energy while deleting a short consonant that distinguishes a number; conversely, a more aggressive filter can improve subjective comfort yet score worse against a clean waveform. For this task, SI-SDR should therefore be reported alongside word-level ASR, named-entity accuracy, and human listening checks.

Across constructed noise conditions, Wiener filtering improves SI-SDR by 0.64-0.70 dB for the three broadband noisy clips and by 0.51 dB for telephone-band speech with added noise; on clean telephone-band input it is neutral. Un-gated DeepFilterNet3 loses 7.38, 4.09, and 1.55 dB on the 15, 10, and 5 dB broadband-noise cases respectively, and 2.42 and 5.19 dB on clean and noisy telephone-band cases. Smart gating avoids the largest neural losses, but its reported deltas are -0.69, +0.26, +0.35, -0.07, and -0.10 dB. Thus the supported conclusion is that gating limits unnecessary processing; the results do not show that gating consistently improves SI-SDR.

5.6 Runtime and causal scope

A frame is 80 ms long, so the reported 1.84-3.01 ms front-end execution time yields RTF = processing time / frame duration = 0.023-0.038. This is equivalent to about 26-43 frame-processing times per unit of audio duration under the reported hardware and configuration. This timing covers the signal-processing front end only. Separately, the 25.47 s source clip is reported to take 21.97 s through the broader offline preprocessing pipeline, which includes ASR, alignment, translation/checking, and file-level operations. These timings refer to different workloads and must not be combined as if they were a single end-to-end live-translation latency.

6 Discussion and Limitations

The experiments support a narrow but useful finding: denoising strength should depend on the acoustic regime. A neural enhancer trained for wideband speech can damage already narrowband telephone speech, while a classical Wiener filter gives modest SI-SDR gains on the constructed noisy copies. Smart gating is a practical protection against unnecessary enhancement, not yet a demonstrated universal optimum. In rows labelled “bypass” in the results log, SI-SDR sometimes shifts slightly from the input; exact waveform comparisons should be checked before claiming bit-identical bypass.

The evidence base is small: one main recording supplies all five signal-processing conditions, only 30 boundaries are manually audited, and the entity checker’s reported 100% pass rate across six clips is not a reliable estimate of general accuracy. Adversarial cases already show gaps for prefix/morphological negation, lexical paraphrases, and repairs without explicit English markers. The full end-to-end speech-to-speech model, listener latency, translation quality, and output-audio naturalness are not benchmarked here. The proposed 200-hour corpus and PEG-SH strategy remain to be implemented and evaluated.

A further issue is the relationship between the gate label and the measured waveform. RESULTS.md describes some rows as bypass operations, yet their reported SI-SDR changes are non-zero. The changes are small in several rows, but a true sample-identical bypass should leave a waveform metric unchanged after equivalent delay alignment. Possible explanations include resampling, gain/level handling, frame-boundary processing, or metric alignment details; the supplied report alone does not resolve which explanation applies. The safe claim is therefore “gated processing stayed near the input SI-SDR,” not “bypass was proven bit-identical.” Exact waveform comparison is needed to distinguish these cases.

The constructed degradations support controlled comparisons, but they are not substitutes for naturally occurring mobile-network audio. Babble, fan noise, band limitation, 8 kHz resampling, and mu-law quantization each model specific impairments; their combination does not cover packet loss, codec variability, reverberation, microphone placement, or changing background speakers in a real call. Performance can also depend on the chosen clean reference and noise mixture. The test set should expand to multiple speakers, dialects, speaking rates, microphones, and independently recorded telephone clips before the architecture is characterized as robust.

A reproducible study should record the gate decision and signal path for every frame, preserve exact frame timestamps, and evaluate whether state is continuous at chunk boundaries. For causal claims, the implementation should be audited for hidden future context in resampling, overlap-add, the neural enhancer, and VAD hangover. A component can process chunks in a loop yet still use lookahead internally; code inspection and controlled impulse/onset tests are needed to establish a strict zero-future-lookahead property. The current report describes a causal design, while independent verification of that property remains outside this document.

6.1 Deployment implications

An 80 ms frame contract gives a fixed input cadence, but it does not by itself guarantee an 80 ms end-to-end response. A live system also pays for capture buffering, resampling, VAD decisions, ASR stabilization, translation decoding, text-to-speech startup, and transport to the listener. The measured 1.84-3.01 ms compute time leaves room in the frame budget for additional stages, but the complete pipeline must still be timed as a single stream. First-audio latency, steady-state throughput, interruption behavior, and final completion lag should be recorded separately. A system that is fast after a translation has been committed can still feel slow if it waits too long before its first useful word.

The reported design also includes a 240 ms VAD hangover to bridge short pauses. This is a reasonable protection against clipped codas and hesitation structure, but it is a temporal policy and should be included in latency accounting. Similarly, a stateful resampler and overlap-add denoiser should be evaluated at frame boundaries, because a chunk-wise Python call does not alone prove that no future samples are consumed or that no boundary artifacts are introduced. An impulse, short plosive, and silence-to-speech transition test can help verify state continuity and onset preservation. These checks are proposed next steps, not results reported in the current benchmark.

6.2 Prioritized validation plan

Three follow-up tests would make the claims more defensible. First, compare the exact input and output waveforms whenever the SmartCausalGate reports a bypass; publish per-condition hashes or maximum absolute error so a bypass is distinguishable from an active filter. Second, extend the degradation test to multiple speakers and independently recorded cellular clips, keeping source speakers separated between tuning and held-out evaluation. Report SI-SDR alongside intelligibility and entity-level transcription, since a waveform metric alone does not establish that a number or negation remains correct. Third, evaluate the full streaming translation system end to end with audio-emission timestamps, translation quality, entity accuracy, and latency percentiles. Front-end frame time, offline preprocessing time, and listener-perceived translation latency should be separate metrics.

A low-cost dataset-mix ablation proposed in the original submission uses a 5-hour subset and 500 update steps: compare 100% synthetic training pairs with a mixture of 60% synthetic and 40% real IndicVoices speech, applying codec augmentation on the fly. A separate 50-sentence stress set should cover compound Hindi numerals, lakh/crore amounts, names, negation, and repairs. Finally, compare fixed wait-3 against PEG-SH on the same audio; measure entity accuracy against latency (including LAAL) and inspect every case where a hold changes the committed spoken value. These are recommended experiments, not completed results.

For each run, freeze the model checkpoint, decoding parameters, audio resampling path, and random seeds. Keep speaker-disjoint training, tuning, and test sets. Report confidence intervals or bootstrap ranges for word error rate and entity accuracy rather than a single aggregate, and publish per-clip results so a small number of difficult repairs cannot be hidden by longer easy utterances. For denoising, report paired SI-SDR deltas with sample counts and include segment-level results for critical entity spans. For streaming translation, report median and 95th-percentile first-audio latency, average lagging or a clearly defined alternative, and total completion lag. A latency claim must specify whether it is measured from microphone frame arrival to translated-audio emission, not simply the neural component runtime.

An evaluation matrix should compare at least: unprocessed input, causal Wiener filtering, un-gated DeepFilterNet3, and Smart-Gated processing. Run every method on identical aligned versions of each test clip. For entity preservation, mark the reference spans for names, addresses, prices, negation, and repair-final values before decoding. Score literal output for faithful repair retention and fluent output for correct final-intent resolution; do not use only exact whole-sentence string matching. Human raters should separately assess intelligibility, semantic correctness, and whether a translation incorrectly commits to a value later corrected by the speaker. These are proposed reporting safeguards, not additional experiments already performed.

6.3 Reproducibility

The supplied RESULTS.md records environment versions, degradation recipes, the SI-SDR definition, and Windows commands for scripts/generate_degraded_clips.py and scripts/eval_part_e.py. The report names output plots and alignment files, but those scripts, raw audio, and plot files were not independently executed or inspected as part of this restructuring. A reproducible release should archive the exact source clip, generated degraded files, per-frame logs, metric script version, gate decisions, and plot outputs together. This is particularly important because the table includes small SI-SDR changes in some rows labelled as bypass.

The recorded runtime environment is Windows 11 Home Single Language, Python 3.12.10, PyTorch 2.14.1+cu130, Torchaudio 2.11.0+cu130, and CUDA 13.0, with an Intel Core i7-14650HX CPU and an NVIDIA RTX 5060 Laptop GPU with 8 GB VRAM. RESULTS.md gives commands to generate degraded clips, run the Part E evaluator, and invoke the interview pipeline. These details support an attempted reproduction, but do not alone prove that all local paths, files, dependencies, and outputs are available to another researcher.

Before release, the artifact bundle should include the clean source waveform, degraded variants, alignment labels, environment lockfile, source code revision, full console logs, and the two named plots. A short README should map every table cell to the script and output file that generated it. Where manual alignment labels are used, retain the word-level timestamps and provide the annotation convention; report the annotator count and adjudication process if additional auditors are recruited. The 30-boundary audit should be expanded with a held-out sample and confidence intervals.

Appendix A. Reproducibility and evaluation protocol

A.1 Reported artifact paths

The results note identifies the featured audio as `data/audio/clip_1_24k.wav`, with aligned labels at `data/alignments/clip_1_audacity.txt`. It names `scripts/generate_degraded_clips.py` for the five altered conditions and `scripts/eval_part_e.py` for the 80 ms causal evaluation and plots. The named plot outputs are `data/part_e_plots/fig1_sisdr_comparison.png` and `data/part_e_plots/fig2_where_denoising_hurts.png`. File paths in the submitted note are local Windows paths; their presence and contents were not independently confirmed during this restructuring.

A.2 Reproduction sequence

The prescribed PowerShell sequence sets UTF-8 output, runs the degradation generator, runs the evaluator, and then invokes the interview pipeline with the front end enabled. In command form, the core sequence is: `python scripts/generate_degraded_clips.py`; `python -u scripts/eval_part_e.py`; and `python scripts/run_live_interview.py --audio data/raw/clip_1.wav --run_sp_frontend`. The actual environment-specific executable path in RESULTS.md is `.venv\Scripts\python.exe`. A fresh run should preserve stdout/stderr and machine-readable outputs rather than only the final Markdown table.

A.3 Minimum result record

For each condition, archive the clean-reference hash, degraded-input hash, processed-output hash, source sample rate, target sample rate, filter/gate decision, measured processing time, delay-alignment offset, input SI-SDR, output SI-SDR, and SI-SDR change. For every bypass decision, calculate maximum absolute sample difference and root-mean-square difference between input and output after accounting for any documented resampling. This makes it possible to determine whether a reported small change comes from true bypass, resampling, gain handling, or metric alignment. Also include the 80 ms frame count and any dropped or repeated frames.

A.4 Proposed held-out tests

A compact follow-up evaluation can use three independently recorded telephone clips per speaker for development and a speaker-disjoint held-out test set, with clean and noisy versions derived from the same source utterances. The stress set should oversample numbers and repairs: compound numerals, address numbers, rupee amounts in lakh/crore units, names that resemble common words, explicit and morphological negation, and interrupted corrections. Evaluate the model with and without denoising on the same samples. Report signal metrics and ASR/entity outcomes together, and include every case in which the final price, person, address, or polarity changes. This protocol is proposed for future work; the reported five-condition benchmark remains a single-source-clip study.

A.5 Interpretation boundary

The measured front-end runtime, the 21.97 s file-level preprocessing time, and the eventual end-to-end streaming latency are distinct quantities. The first describes processing time for an 80 ms frame; the second describes a pipeline run on one finite clip; the third must be measured from incoming audio to emitted translated speech under streaming conditions. Likewise, SI-SDR measures waveform fidelity against a paired reference and cannot substitute for the preservation of names, numbers, negation, or the resolved meaning of a correction. Claims about translation quality, naturalness, or human-usefulness should wait for dedicated evaluations.

A.6 Proposed error taxonomy

For future evaluation, annotate errors in categories that correspond to the task's high-risk content: (i) entity substitution, such as changing Sector 62 to another address; (ii) numeric magnitude errors, such as confusing 1.20 crore with 1.02 crore or 10 lakh with 8 lakh; (iii) polarity errors involving "not", refusal, or morphological negation; (iv) repair-resolution errors in which the abandoned value remains in fluent output; (v) repair-deletion errors in which a literal output omits a meaningful self-correction; (vi) timing errors where an English word is emitted before its supporting source span; and (vii) acoustic errors caused by clipping, over-attenuation, or frame-boundary artifacts. The categories should be mutually defined before annotation, with ambiguous examples adjudicated and included in the supplementary error log.

A literal translation and a fluent translation should be scored against different reference expectations. Literal output is expected to preserve repetitions, hesitations, and corrections in a readable form; fluent output is expected to resolve the speaker's final intention without retaining an abandoned numeric value as the answer. A single exact-match score can unfairly penalize a good literal rendering or reward a fluent output that drops the correction. Report separate scores for content preservation, final-intent resolution, and timing of commitment. For each category, publish both the numerator and denominator—for example, the number of correct final prices over all clips containing a price—so small test sets are not obscured by an aggregate percentage.

A.7 Suggested reporting table

Evaluation item	Required report
Alignment	Number of audited boundaries, annotator convention, MAE before/after clamp, and held-out error distribution.
Audio quality	Input/output SI-SDR, paired delta, sample count, delay offset, and intelligibility checks for each condition.
Entities and repairs	Correct names, addresses, amounts, negations, and repair-final values, with counts and failure examples.
Streaming latency	First emitted audio, median and 95th-percentile latency, steady-state RTF, and end-of-turn completion lag.
Causality	Lookahead audit, state continuity, frame drops, and maximum input-output difference for bypass paths.

Table 2. Recommended future reporting structure; this table is a protocol proposal, not a report of additional experiments.

A.8 Pre-submission checks

Before the competition submission, verify that every reported measurement can be regenerated from a clean environment; record the exact code revision and model identifiers; confirm that each plot matches the table values; inspect the PDF and audio assets for missing files; and ensure that proposed dataset hours and latency-gating ideas are labelled as future work until implemented. The final experimental summary should use consistent terms for "front-end runtime", "offline preprocessing time", and "end-to-end streaming latency". It should also distinguish measured values, values computed from reported measurements, and qualitative interpretations. These checks would make the evidence easier for reviewers to audit without overstating what has been demonstrated.

7 Conclusion

This compact submission presents a causal 80 ms audio front end and a repair-aware data/alignment design for code-mixed Hindi-to-English streaming translation. Reported tests show a 65.3% drop in word-boundary MAE after RMS clamping, low front-end frame time, and a clear failure case for un-gated neural denoising on telephone-band audio. The next step is broader multi-speaker evaluation, exact waveform/log verification, and a measured end-to-end streaming S2ST benchmark against the unchanged base checkpoint.

References

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Source note: Quantitative measurements are based on the supplied original mini-PS PDF and RESULTS.md. Report structure and academic style are inspired by the supplied ACL paper; no authorship or experimental results from that paper are reused.